

Product Requirements Document (PRD)

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1. Product Overview

StudyTrack is a mobile-first web application that helps students plan, prioritize, and complete academic work. Unlike generic to-do apps, StudyTrack organizes tasks by course and due date, surfaces the week's workload at a glance, and rewards consistent completion with visible progress. Its differentiators are one-tap import from Canvas and Google Calendar, so a student's real deadlines appear automatically, and an AI planning assistant (powered by the Claude API) that turns those deadlines into a personalized weekly study schedule. The MVP targets undergraduate students and is designed to be fully set up in under five minutes.

2. Business Goal

Validate that a course-aware task manager can win and retain students against generic productivity tools. The MVP aims to acquire 10,000 registered users within two academic terms, achieve 35% 30-day retention, and establish the engagement baseline (weekly active use, on-time completion rate, import adoption) needed to justify a premium tier with analytics and grade-weight prioritization.

3. User Personas

Persona 1: Maya, the Overloaded Undergrad

- 19, sophomore taking 5 courses plus a part-time job. Tracks deadlines across three apps, misses small assignments, and crams before exams. Needs one place showing what is due this week and what to start now.

Persona 2: Jordan, the Ambitious High Schooler

- 17, junior balancing AP classes, SAT prep, and college applications. His paper planner doesn't remind him of anything. Needs simple recurring tasks that keep long-term goals from slipping.

Persona 3: Priya, the Part-Time Grad Student

- 28, works full-time with two evening courses and limited study windows. Needs to see the workload by week to schedule study sessions in advance.

4. Features (MVP Scope)

- **Account management:** Email/password sign-up, login, and logout.
- **Course management:** Create, view, edit, and delete courses (name, color, term).
- **Task management:** Create, view, edit, and delete tasks with title, description, course, due date, priority, and estimated effort.
- **Planning views:** "Today" and "This Week" views sorted by due date and priority; overdue tasks clearly flagged.
- **Tracking & progress:** One-tap task completion with completion history and a simple progress dashboard (tasks completed per week, on-time rate, current streak).
- **Calendar & LMS import:** Import assignments and events from Canvas (via iCal feed) and Google Calendar (via OAuth). Imported deadlines are auto-converted into tasks,

mapped to the right course, refreshed daily, and de-duplicated. This is the core differentiator: students see their real deadlines without manual entry.

- **AI planning assistant:** An “Ask Claude” feature powered by the Claude API. When a student asks how to plan more efficiently, Claude analyzes their open tasks (due dates, priorities, effort) and generates a personalized, editable weekly study schedule. Limited plans per week on the free tier; unlimited on paid.
- **Explicitly out of scope:** Out of scope for MVP: push notifications, two-way calendar sync (import only), collaboration/shared tasks, and native mobile apps. These are fast-follow candidates.

5. Functional Requirements

- FR-1. Users can register with name, email, and password; the system rejects duplicate emails and weak passwords (minimum 8 characters).
- FR-2. Users can log in with valid credentials and log out; invalid credentials return a clear error without revealing which field was wrong.
- FR-3. Authenticated users can create a task with a required title and due date; description, course, priority, and effort are optional.
- FR-4. Users can edit and delete only their own tasks and courses; all destructive actions require confirmation.
- FR-5. The task list defaults to a “This Week” view sorted by due date, with filters for course, priority, and completion status.
- FR-6. Marking a task complete records a timestamped completion entry; tasks can be uncompleted, which removes the entry.
- FR-7. The dashboard displays tasks completed this week, on-time completion percentage, and current daily completion streak.
- FR-8. Users can connect Google Calendar via OAuth and Canvas via an iCal feed URL; imported items become tasks with source labels and are refreshed at least daily.
- FR-9. Re-importing the same feed never creates duplicate tasks; users can disconnect a source at any time, choosing to keep or remove its imported tasks.
- FR-10. Users can request an AI-generated weekly plan; the system sends only the user’s open tasks (titles, due dates, priorities, effort) to the Claude API and returns a schedule the user can edit, regenerate, or dismiss. Requests are rate-limited per tier.
- FR-11. All API endpoints validate input server-side and return meaningful HTTP status codes and error messages.

6. Non-Functional Requirements

- **Performance:** Core pages load in under 2 seconds on a mid-range phone over 4G; API responses under 300 ms at MVP scale.
- **Security:** Passwords hashed (bcrypt or equivalent); all traffic over HTTPS; users can only access their own data.
- **Usability & accessibility:** Fully responsive layout, designed mobile-first; meets WCAG 2.1 AA contrast and touch-target guidance.
- **Reliability:** 99% uptime target during the academic term; graceful error states when the backend is unreachable.
- **Scalability:** Architecture supports 10,000 users without redesign (stateless API, indexed PostgreSQL queries).

7. Assumptions

- Students will tolerate a web app (no native app) for the MVP if it is fast and mobile-responsive.
- Most students' institutions expose Canvas iCal feeds, and students are willing to connect their Google account; manual entry remains available as a fallback.
- One-way import (calendar → StudyTrack) is sufficient for MVP; two-way sync can follow.
- Email/password authentication is sufficient for MVP; social login can follow.
- The primary device is a smartphone; the desktop is secondary.

8. Risks

Risk	Impact	Mitigation
Students abandon the app after initial setup (novelty wears off)	Low retention undermines business goals	Streaks, weekly progress summary, fast task entry (under 10 seconds)
Generic tools (Notion templates, Google Tasks) are "good enough."	Weak differentiation slows adoption	Lead with automatic Canvas/Google Calendar import and five-minute setup vs. hours of configuration in generic tools
Import breaks or imports messy data (Canvas feed formats vary; OAuth friction)	The core differentiator fails the first impression	Robust feed parsing with preview-before-import; manual entry always available as fallback
AI plans are generic, wrong, or costly at scale (per-request API costs)	Feature disappoints; margins erode	Prompt includes real task data for personalization; rate limits per tier; plans clearly labeled as editable suggestions
Security breach exposing student data	Loss of trust; possible legal exposure	Hashing, HTTPS, input validation, least-privilege data access from day one

9. Success Metrics

- **Activation:** Onboarding completion rate above 80% (account created → first task added); 50%+ of new users connect Canvas or Google Calendar in week one.
- **Engagement:** 40% weekly active users during the term; 5+ tasks created per active user per week.
- **Retention:** 35% 30-day retention among activated users.
- **Outcome:** 70%+ of tasks completed on or before the due date.
- **Quality:** Crash-free sessions above 99.5%; median API latency under 300 ms.